

Rob 501 Fall 2014
Lecture 13
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Weighted Least Squares

Let Q be an $n \times n$ positive definite matrix ($Q > 0$) and let the inner product on $\mathbb{R}^n$ be

$$\langle x, y \rangle = x^\top Q y.$$

We re-do $A\alpha = b$, where $A = n \times m, n \geq m, \text{rank}(A) = m, \alpha \in \mathbb{R}^m$, and $b \in \mathbb{R}^n$. We want to seek $\hat{\alpha}$ such that

$$\|A\hat{\alpha} - b\| = \min_{\alpha \in \mathbb{R}^m} \|A\alpha - b\|$$

where $\|x\| = \langle x, x \rangle^{\frac{1}{2}} = (x^\top Q x)^{\frac{1}{2}}$ and $Q > 0$.

Solution: $\mathcal{X} = \mathbb{R}^n, \mathcal{F} = \mathbb{R}, \langle x, y \rangle = x^\top Q y$

Write $A = [A_1 \mid A_2 \mid \cdots \mid A_m]$

Normal Equations:

$$\begin{aligned}\hat{x} &= \hat{\alpha}_1 A_1 + \hat{\alpha}_2 A_2 + \cdots + \hat{\alpha}_m A_m \\ G^\top \hat{\alpha} &= \beta, \text{ with } G = G^\top \\ [G^\top]_{ij} &= [G]_{ij} = \langle A_i, A_j \rangle = A_i^\top Q A_j = [A^\top Q A]_{ij} \\ \beta_i &= \langle b, A_i \rangle = b^\top Q A_i = A_i^\top Q b = [A^\top Q b]_i.\end{aligned}$$

$$\therefore A^\top Q A \hat{\alpha} = A^\top Q b.$$

Since $A^\top Q A$ is invertible by $\text{rank}(A) = m$, we can conclude that

$$\underline{\hat{\alpha} = (A^\top Q A)^{-1} A^\top Q b.}$$

Recursive Least Squares

Model:

$$y_i = C_i x + e_i, \quad i = 1, 2, 3, \dots$$

$$C_i \in \mathbb{R}^{m \times n}$$

i = time index

x = an unknown constant vector $\in \mathbb{R}^n$

y_i = measurements $\in \mathbb{R}^m$

e_i = model "mismatch" $\in \mathbb{R}^m$

Objective 1: Compute a least squared error estimate of x at time k , using all available data at time k , $(y_1, \dots, y_k)$!

Objective 2: Discover a computationally attractive form for the answer.

Solution:

$$\begin{aligned}\hat{x}_k &:= \operatorname{argmin}_{x \in \mathbb{R}^n} \left(\sum_{i=1}^k (y_i - C_i x)^\top S_i (y_i - C_i x) \right) \\ &= \operatorname{argmin}_{x \in \mathbb{R}^n} \left(\sum_{i=1}^k e_i^\top S_i e_i \right)\end{aligned}$$

where $S_i = m \times m$ positive definite matrix. ($S_i > 0$ for all time index i)

Batch Solution:

$$Y_k = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_k \end{bmatrix}, A_k = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_k \end{bmatrix}, E_k = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_k \end{bmatrix}$$

$$R_k = \begin{bmatrix} S_1 & & & \\ & S_2 & & \\ & & \ddots & \\ & \mathbf{0} & & S_k \end{bmatrix} = \text{diag}(S_1, S_2, \dots, S_k) > 0$$

$$Y_k = A_k x + E_k, \text{ [model for } 1 \leq i \leq k]$$

$$\|Y_k - A_k x\|^2 = \|E_k\|^2 := E_k^\top R_k E_k$$

Since $\hat{x}_k$ is the value minimizing the error $\|E_k\|$, which is the unexplained part of the model,

$$\hat{x}_k = \underset{x \in \mathbb{R}^n}{\text{argmin}} \|E_k\| = \underset{x \in \mathbb{R}^n}{\text{argmin}} \|Y_k - A_k x\|,$$

which satisfies the Normal Equations $(A_k^\top R_k A_k) \hat{x}_k = A_k^\top R_k Y_k$.

$\therefore \underline{\hat{x}_k = (A_k^\top R_k A_k)^{-1} A_k^\top R_k Y_k}$, which is called a Batch Solution.

Drawback: $A_k = km \times n$ matrix, and grows at each step!

Solution: Find a recursive means to compute $\hat{x}_{k+1}$ in terms of $\hat{x}_k$ and the new measurement y_{k+1} !

Normal equations at time k , $(A_k^\top R_k A_k) \hat{x}_k = A_k^\top R_k Y_k$, is equivalent to

$$\left(\sum_{i=1}^k C_i^\top S_i C_i \right) \hat{x}_k = \sum_{i=1}^k C_i^\top S_i y_i.$$

We define

$$Q_k = \sum_{i=1}^k C_i^\top S_i C_i$$

so that

$$Q_{k+1} = Q_k + C_{k+1}^\top S_{k+1} C_{k+1}.$$

At time $k + 1$,

$$\underbrace{\left(\sum_{i=1}^{k+1} C_i^\top S_i C_i \right)}_{Q_{k+1}} \hat{x}_{k+1} = \sum_{i=1}^{k+1} C_i^\top S_i y_i$$

or

$$Q_{k+1} \hat{x}_{k+1} = \underbrace{\sum_{i=1}^k C_i^\top S_i y_i}_{Q_k \hat{x}_k} + C_{k+1}^\top S_{k+1} y_{k+1}.$$

$$\underline{\therefore Q_{k+1} \hat{x}_{k+1} = Q_k \hat{x}_k + C_{k+1}^\top S_{k+1} y_{k+1}}$$

Good start on recursion! Estimate at time $k + 1$ expressed as a linear combination of the estimate at time k and the latest measurement at time $k + 1$.

Continuing,

$$\hat{x}_{k+1} = Q_{k+1}^{-1} [Q_k \hat{x}_k + C_{k+1}^\top S_{k+1} y_{k+1}].$$

Because

$$Q_k = Q_{k+1} - C_{k+1}^\top S_{k+1} C_{k+1},$$

we have

$$\hat{x}_{k+1} = \hat{x}_k + \underbrace{Q_{k+1}^{-1} C_{k+1}^\top S_{k+1}}_{\text{Kalman gain}} \underbrace{(y_{k+1} - C_{k+1} \hat{x}_k)}_{\text{Innovations}}.$$

Innovations $y_{k+1} - C_{k+1} \hat{x}_k$ = measurement at time $k + 1$ minus the "predicted" value of the measurement = "new information".

In a real-time implementation, computing the inverse of Q_{k+1} can be time consuming. An attractive alternative can be obtained by applying the Matrix Inversion Lemma:

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B (DA^{-1}B + C^{-1})^{-1} DA^{-1}$$

Now, following the substitution rule as shown below,

$$A \leftrightarrow Q_k \quad B \leftrightarrow C_{k+1}^\top \quad C \leftrightarrow S_{k+1} \quad D \leftrightarrow C_{k+1},$$

we can obtain that

$$\begin{aligned} Q_{k+1}^{-1} &= (Q_k + C_k^\top S_{k+1} C_{k+1})^{-1} \\ &= Q_k^{-1} - Q_k^{-1} C_{k+1}^\top [C_{k+1} Q_k^{-1} C_{k+1}^\top + S_{k+1}^{-1}]^{-1} C_{k+1} Q_k^{-1}, \end{aligned}$$

which is a recursion for Q_k^{-1} !

Upon defining

$$P_k = Q_k^{-1},$$

we have

$$P_{k+1} = P_k - P_k C_{k+1}^\top [C_{k+1} P_k C_{k+1}^\top + S_{k+1}^{-1}]^{-1} C_{k+1} P_k$$

We note that we are now inverting a matrix that is $m \times m$, instead of one that is $n \times n$. Typically, $n > m$, sometimes by a lot!